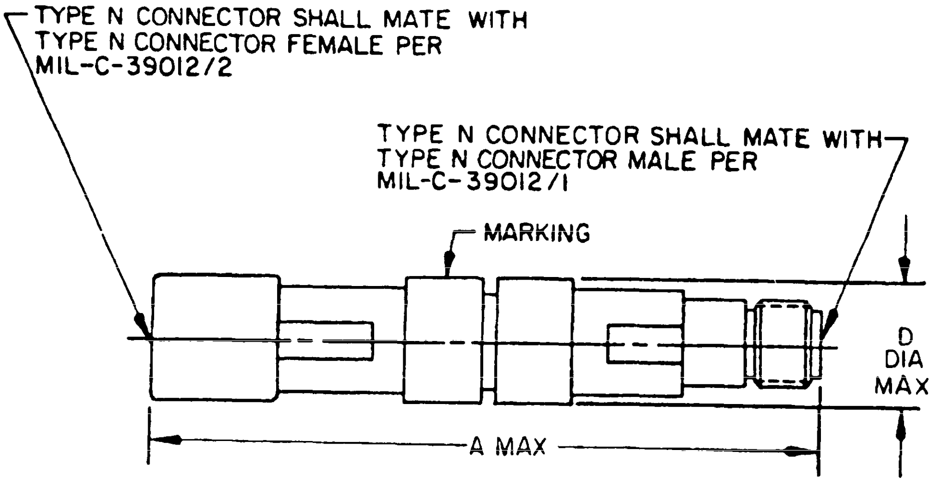
MIL-A-3933/9B
24 April 1985
SUPERSEDING
MIL-A-3933/9A
28 March 1973

MILITARY SPECIFICATION SHEET

ATTENUATORS, FIXED, COAXIAL LINE (SERIES N), FREQUENCY RANGE: DC TO 4.5 GHz, CLASS III, LOW POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the attenuators described herein shall consist of this specification and the latest issue of MIL-A-3933.


TYPE N CONNECTOR SHALL MATE WITH TYPE N CONNECTOR FEMALE PER MIL-C-39012/2
TYPE N CONNECTOR SHALL MATE WITH TYPE N CONNECTOR MALE PER MIL-C-39012/1
MARKING
D DIA MAX
A MAX

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration.

ENGINEERING DATA:

Class III.

Operating frequency range: DC to 4.5 gigahertz (GHz).

Test frequencies for VSWR and attenuation measurements: DC, 1, 2, 3 and 4.5 GHz.

Basic test frequency: 2.0 GHz.

Operating temperature range: -55°C to +125°C.

Maximum weight: See table I.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Nominal impedance: 50 ohms.

Attenuation: See table I.

Power:
Average - watts (continuously) maximum, see table I.
Peak - 2 kilowatts minimum for a duty cycle of 5×10^{-4}; maximum pulse duration 5 microseconds.

VSWR: maximum for both ends, see table I.

Part number: M3933/9- (and dash number from table I).

TABLE I. Electrical characteristics.

Dash number	Attenuation dB Nominal	Attenuation dB Deviation DC to 4.5 GHz	AV power	VSWR	Maximum weight (oz)	Dimensions A	Dimensions D
04 N	1	±0.2	4	1.35:1	2.8	2.75	0.875
S						(69.9)	(22.23)
05 N	2	±0.2	4	1.35:1	2.8	2.75	0.875
S						(69.9)	(22.23)
06 N	3	±0.2	4	1.35:1	2.8	2.75	0.875
S						(69.9)	(22.23)
07 N	4	±0.2	3	1.35:1	2.8	2.75	0.875
S						(69.9)	(22.23)
08 N	5	±0.2	3	1.35:1	2.8	2.75	0.875
S						(69.9)	(22.23)
09 N	6	±0.3	3	1.35:1	2.8	2.75	0.875
S						(69.9)	(22.23)
10 N	8	±0.3	2	1.25:1	2.8	2.75	0.875
S						(69.9)	(23.23)
11 N	10	±0.3	2	1.25:1	2.8	2.75	0.875
S						(69.9)	(22.23)
12 N	12	±0.4	2	1.25:1	2.8	2.75	0.875
S						(69.9)	(22.23)
13 N	15	±0.4	2	1.25:1	2.8	2.75	0.875
S						(69.9)	(22.23)
14 N	20	±0.4	2	1.25:1	2.8	2.75	0.875
S						(69.9)	(22.23)
15 N	25	±0.6	2	1.25:1	2.8	2.75	0.875
S						(69.9)	(22.23)
16 N	30	±0.6	2	1.25:1	2.8	2.75	0.875
S						(69.9)	(22.23)
17 N	50	±1.0	2	1.25:1	3.2	4.0	0.875
S						(101.6)	(22.23)
18 N	60	±1.2	2	1.25:1	3.2	4.0	0.875
S						(101.6)	(22.23)

NOTE: Attenuators covered by this specification sheet are substitutable for the manufacturer's part numbers specified in table II. This information in no way implies that the manufacturer's part number or NSN is suitable as a substitute for the military part number.

TABLE II. Substitution data.

Military part number M3933/9-	Manufacturer's code	Manufacturer's part number	NSN
04	00929	ABJ 71	5985-01-054-4624
05	00929	AA-02N	5985-00-009-1475
05	00929	AB-02N	---
05	00929	ABJ 75	5985-01-054-4265
05	23042	FP50-2N	5985-00-054-4265
06	00929	AB-3N	5985-00-957-4814
06	80058	CN-1298/U	---
06	07421	318-200-103	5985-00-957-4814
06	23042	FP50-3N	5985-00-239-9435
07	00929	AA-04N	5985-00-559-1365
08	00929	AA-05N	5985-00-927-3101
09	00929	AB-06N	5985-00-856-6293
09	00929	AA-06N	5985-00-507-8934
09	23042	FP50-6N	5985-01-140-8214
09	00929	AE-06N	5985-00-137-8124
10	00929	AB-08N	5985-00-983-4940
11	00929	AA-10N	5985-00-082-0603
11	00929	AB-10N	5985-00-835-3934
11	23042	FP50-10N	5985-01-038-2655
12	00929	AA-12N	5985-00-835-9060
12	00929	AB-12N	5985-00-931-9671
13	18203	A415N	5985-01-061-8240
13	00929	AB-15N	5985-00-810-2464
14	00929	AA-20N	5985-00-794-8765
14	00929	AE-20N	5985-00-892-9451
15	00929	AB-25N	5985-01-024-7589
16	00929	AB-30N	5985-00-764-8502
16	00929	AA-30N	5985-00-853-2893
16	00929	AE-30N	5985-00-225-8621
17	00929	AB-50N	5985-00-408-3797
17	00929	AA-50N	5985-00-870-0795
18	00929	AB-60N	5985-00-924-4503

Supersession data: See table III.

TABLE III. Supersession data.

Part number M3933/9	Supersedes M3933/
-06	9-1
-16	9-2
-17	9-3

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy -
Air Force - 11, 99
DLA - ES

User activities:
Army - SM
Navy - MC, SH, AS

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5935-0993-7)

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☐ OTHER (Specify): ______

5. PROBLEM AREAS

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b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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